

Hamza Alshamy

 GitHub |  LinkedIn |  Website |  ha2486@nyu.edu

EDUCATION

New York University

Sep 2024 – May 2026

M.S. in Data Science

- Research area: Computational Social Science

B.A. in Economics

May 2024

- Minors: Data Science and Mathematics

RESEARCH EXPERIENCE

Center for Conflict and Cooperation

Aug 2025 – Present

GRADUATE RESEARCH ASSISTANT

PI: [Dr. Jay Van Bavel](#); Supervisor: [Dr. Laura Globig](#)

- Developing NLP classifiers and experimental methods to evaluate how social incentives and AI interventions reduce the spread of misinformation and harmful content online.

Centre for Human-Inspired AI, University of Cambridge

Jul 2024 – Present

VISITING RESEARCH ASSISTANT

PI: [Dr. Umang Bhatt](#)

- Designing a longitudinal experiment to evaluate how AI assistance affects skill development, incentives, and promotion dynamics among knowledge-workers.

Venn Research Group (*Independent*)

Nov 2023 – Present

RESEARCH LEAD

Scientific Advisor: [Dr. Pascal Wallisch](#)

- Leading interdisciplinary research by utilizing Data Science to study complex social systems and emergent behaviors across various fields.

New York University

Dec 2023 – May 2024

LEAD QUANTITATIVE RESEARCHER (CO-PI)

Faculty Advisor: [Dr. Gerald McIntyre](#)

- Built and cleaned a 100-country World Bank panel dataset integrating income, demographic, and unemployment indicators.
- Estimated fixed and random-effects models that isolated demographic drivers of GDP growth and unemployment.

PUBLICATIONS

- [1] **What NLP says about politics: Longitudinal sentiment analysis of U.S. presidential debates (1960–2024)**
Draft in preparation for Journal of Computational Social Science (Draft PDF )
Hamza Alshamy, Alexander Pegot-Ogier
- [2] **TimeGraph: A computer vision pipeline for quantifying territorial dynamics from animated historical maps**
Work in progress
Hamza Alshamy, Advay Mirsha, Isaiah Woram, Charlie Xia, Pascal Wallisch
- [3] **Saudi demographic economics: Population aging and drivers of declining fertility rates**
King Faisal Center for Research and Islamic Studies, Masarat, June 2025 
Hamza Alshamy

	[4] Demographic economics: The implications and consequences of Japan’s aging population on labor productivity, immigration, and pension funds <i>Annual Undergraduate Research Conference, NYU, Oral Presentation, May 2024 (Best Social Science Research Award)</i>  Hamza Alshamy, Anviti Swaraj, Nikko Elyassi																		
TEACHING EXPERIENCE	New York University, Graduate Teaching Assistant <i>Fall 2025</i> DS-UA 201: Causal Inference (Instructor: Dr. Sidharth Sah) DS-GA 1001: Introduction to Data Science (Instructor: Dr. Pascal Wallisch) – Master’s <i>Spring 2025</i> CSCI-UA 473: Fundamentals of Machine Learning (Instructor: Dr. Pascal Wallisch) DS-UA 112: Principles of Data Science II (Instructor: Dr. Pascal Wallisch)																		
FELLOWSHIPS, FUNDING,& AWARDS	<table border="0"> <tr> <td>King Abdullah Full Merit Scholarship – Graduate Studies</td> <td>2024 – 2026</td> </tr> <tr> <td>Frédéric Bastiat Fellow </td> <td>2024 – 2025</td> </tr> <tr> <td><i>The Mercatus Center at George Mason University</i></td> <td></td> </tr> <tr> <td>Best Social Science Research Award </td> <td>2024</td> </tr> <tr> <td><i>50th Annual Undergraduate Research Conference, NYU</i></td> <td></td> </tr> <tr> <td>NYU Dean’s Undergraduate Research Fund </td> <td>2024</td> </tr> <tr> <td>Joseph Schumpeter Fellow </td> <td>2021 – 2022</td> </tr> <tr> <td><i>The Mercatus Center at George Mason University</i></td> <td></td> </tr> <tr> <td>King Abdullah Full Merit Scholarship – Undergraduate Studies</td> <td>2020 – 2024</td> </tr> </table>	King Abdullah Full Merit Scholarship – Graduate Studies	2024 – 2026	Frédéric Bastiat Fellow 	2024 – 2025	<i>The Mercatus Center at George Mason University</i>		Best Social Science Research Award 	2024	<i>50th Annual Undergraduate Research Conference, NYU</i>		NYU Dean’s Undergraduate Research Fund 	2024	Joseph Schumpeter Fellow 	2021 – 2022	<i>The Mercatus Center at George Mason University</i>		King Abdullah Full Merit Scholarship – Undergraduate Studies	2020 – 2024
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SELECT PROJECTS	A more exhaustive list of my projects can be found on GitHub. Fine-tuning Transformer-based (BERT) Language Model with Parameter-Efficient Methods  Implemented BitFit for efficient deployment of BERT models on sentiment tasks using Hugging Face and Optuna. Reduced trainable parameters by 99.93% (4.4M → 3K) while preserving functional classification performance. A Course in Principles of Data Science: Foundations of Statistics and Machine Learning  Designed and publicly released a <i>14-lecture slide series</i> to support data science education. Covered probability, inference, regression, hypothesis testing, and machine learning to aid students across disciplines.																		
TECHNICAL SKILLS	• Programming Languages: Python, R, SQL • Tools & Frameworks: Git, Hadoop, Spark • ML/DL Libraries: TensorFlow, PyTorch, SciKit-learn, Hugging Face • Applied Skills: NLP, Social Media Analysis, Clustering, Probabilistic Modeling, Visualization																		
SELECT COURSEWORK	Machine Learning (Kyunghyun Cho); Natural Language Understanding (Tal Linzen); Probability and Statistics for Data Science (Carlos Fernández-Granda); Development Economics (William Easterly); History of Economic Thought (Mario Rizzo)																		